

Mike Hall

Staff Observability & Resilience Architect

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Observability and Resilience Architect specializing in illuminating dark telemetry, enterprise OpenTelemetry rollouts, and deep incident root cause triage. Eliminates cross-service blind spots by engineering end-to-end distributed trace propagation across legacy microservices and modern cloud infrastructure.

EXPERIENCE & LEADERSHIP

Associate Director, Staff Engineer

OneMain Financial | Remote | January 2021 - February 2026

Target Focus: Founded and led the OpenTelemetry Working Group, establishing distributed observability across application and integration boundaries and connecting system signals to incident response and safer change.

- **Originations IC Delivery & Instant Disbursement:** Led the Originations Verification squad through consecutive Exceeds Expectations ratings, architecting and shipping an instant loan disbursement pipeline (funding to debit cards in minutes). Appointed Software Architect for the Acquisition Lane, later converting to Associate Director, Staff Engineer.
- **Acquisition Workflow Enablement & Architecture Discovery:** Founded the Acquisition Lane enablement team; mapped the loan acquisition and origination pipeline end-to-end from visit to booked loan (desktop browser to mainframe and back again at each step) and the Affiliate API at the same depth, identifying seven distinct lending funnels with a tracking initiative; architected an automated 5-phase PII remediation deletion engine across 30+ tables.
- **Enterprise Resilience & DynamoDB Session Remediation:** Diagnosed and eliminated a critical session-cookie overflow defect that caused 4% silent traffic loss of digital loan applications during e-signing; architected zero-downtime blue/green migration to DynamoDB session store with zero incidents.
- **Enterprise Trace & Operational Alignment:** Architected distributed telemetry across Rails distributed monoliths, MuleSoft APIs, and IBM mainframe backends; partnered directly with Cybersecurity, EMC, SRE, and Incident Command to align monitoring with end-to-end distributed tracing.
- **Community Enablement & SRE Handoff:** Founded weekly Geekfest@OMF and scaled the enterprise OpenTelemetry Working Group to 40+ cross-lane engineers; mentored and partnered with SRE leads to transition ongoing operational facilitation, establishing a durable community of practice.
- **Applied AI & Innovation Leadership:** Placed in two corporate hackathons building conversational agents with Rasa and a schema inference prototype; pioneered local LLM orchestration and privacy-conscious AI developer tooling at Geekfest@OMF.

Principal Architect

Local AI Orchestration & Developer Runtime | Independent Project | January 2026 - Present

Target Focus: Built telemetry-driven diagnosis MCP server exposing distributed traces and error budgets directly to AI agent runtimes.

- **Bidirectional Knowledge Server:** Built ctx-mcp, an MCP server fronting a PostgreSQL knowledge suite that agents both query and write back to, so methodologies and lessons learned in one session are retrievable in the next rather than re-derived.
- **Telemetry-Driven Diagnosis:** Built o2-mcp, exposing on-demand OpenObserve telemetry (errors, slow spans, failed jobs, distributed traces) as agent-callable tools, so an agent diagnoses a running system from observed behavior instead of inferring from source.
- **Context as a Budgeted Resource:** Designed context-isolated subagents that keep high-volume audit work out of the main conversation's context window, with per-client registration tooling and pre-commit hooks that reject any agent asset that isn't self-describing.

Principal Architect

WWWorkRemote | Independent Project | January 2021 - Present

Target Focus: Built comprehensive error tracking and validation guardrails for continuous document ingestion and model inference.

- **Multi-Source Data Ingestion:** Built a multi-source Rails ingestion pipeline and PostgreSQL storage schema, unifying fragmented job market datasets for structured downstream analysis.
- **Semantic Search & Embeddings:** Implemented vector search embeddings via pgvector, enabling high-precision semantic matching between complex candidate profiles and job requirements.
- **Local-First AI Orchestration:** Designed provider-agnostic LLM orchestration layer (llama.cpp/Ollama with hosted Claude fallback), enforcing local-first inference for sensitive data with per-call model routing.
- **Prompt Injection Defense:** Built a 4-stage guardrails pipeline screening untrusted scraped text via weighted heuristics and instruction-density scoring before model ingestion.
- **Browser Extension & Lifecycle Capture:** Built an MV3 Chrome extension with 16 provider adapters that extracts postings from live ATS pages and captures full application lifecycles.
- **Architectural Constraint Injection:** Injected static analysis rules and architectural constraints into model context windows, preventing drift between generated outputs and security contracts.

Senior Software Developer

ActiveCampaign | Chicago, IL | September 2018 - December 2018

Target Focus: Made previously opaque PHP backend behavior observable through structured query patterns and test coverage, improving change safety and developer feedback loops.

- **Legacy Query Modernization:** Replaced ad-hoc global data access with structured, cacheable query patterns in the PHP backend, enabling unit testability and sub-second query execution.
- **Frontend Verification Gates:** Hardened Ember.js automated test suites across core CRM contact workflows, creating safety gates that enabled non-breaking legacy PHP refactoring.

Software Engineer & Technical Onboarding Lead (Fraud & Taxonomy Systems)

Groupon | Chicago, IL | July 2011 - May 2013

Target Focus: Engineered fraud detection queries in Vertica and conducted exploratory anomaly detection spikes in Hadoop and Clojure.

- **Centralized Taxonomy Service:** Implemented merchant taxonomy service in Java and MySQL, eliminating categorization drift across distributed product teams during hypergrowth.
- **Analytical Fraud Detection:** Built analytical fraud detection queries in Vertica and conducted exploratory spikes on Hadoop and Clojure for transaction anomaly scoring.
- **Merchant Analytics Pipelines:** Built high-throughput merchant analytics tools in Ruby and CouchDB, surfacing market insights for global sales operations.
- **Global Engineering Enablement:** Redesigned technical onboarding curricula, standardizing engineering practices and shortening time-to-first-commit for 100+ global engineering hires.

SELECTED PRODUCTION PROJECTS

Principal Architect

Local AI Orchestration & Developer Runtime | Independent Project | January 2026 - Present

Built and operate three MCP (Model Context Protocol) servers that expose live system state as callable tools to any MCP client, alongside the Claude Code skills, context-isolated subagents, and commit-time checks that keep agent work bounded and reviewable.

- **Bidirectional Knowledge Server:** Built ctx-mcp, an MCP server fronting a PostgreSQL knowledge suite that agents both query and write back to, so methodologies and lessons learned in one session are retrievable in the next rather than re-derived.
- **Telemetry-Driven Diagnosis:** Built o2-mcp, exposing on-demand OpenObserve telemetry (errors, slow spans, failed jobs, distributed traces) as agent-callable tools, so an agent diagnoses a running system from observed behavior instead of inferring from source.
- **Context as a Budgeted Resource:** Designed context-isolated subagents that keep high-volume audit work out of the main conversation's context window, with per-client registration tooling and pre-commit hooks that reject any agent asset that isn't self-describing.

Principal Engineer & Curator

Technical Conversation Archive | Independent Project | January 2011 - Present

Recorded, and later restored, an archive of 214 technical interviews with practitioners from the Ruby, JVM, and Software Craftsmanship movements, using a local-only AI pipeline to transcribe, structure, and cross-link a longitudinal record of primary-source material into a searchable knowledge graph.

- **Local-Only AI Restoration Pipeline:** Built a Whisper and local-LLM pipeline that transcribes, diarizes, and structures long-form interview audio entirely on-device, keeping a large private media corpus off third-party inference services.
- **Schema-Validated Content Platform:** Runs on a contract-validated data layer: every interview, asset, and transcript is checked against an explicit schema at build time, with referential-integrity checks that fail the build rather than publishing broken data.
- **Semantic Cross-Linking & Taxonomy Generation:** Generates a topic taxonomy, knowledge graph, and semantic cross-links across the corpus, turning an unstructured media archive into navigable, individually indexed pages.
- **Primary-Source Capture:** Conducted the original interviews on-site at GOTO Conference, Software Craftsmanship North America, RailsConf, and WindyCityRails, with practitioners including Dave Thomas, Stuart Halloway, Corey Haines, Sandro Mancuso, and Micah Martin.
- **Digital Archaeology:** Recovered and reconstructed material from defunct platforms and web archives, reconciling incomplete metadata across sources to restore provenance for recordings that would otherwise have been lost.

CORE SKILLS & COMPETENCIES

01 • OpenTelemetry Working Group Leadership & Governance	02 • Distributed Trace Context Propagation (W3C Trace Context)	03 • OTEL Collector Pipeline Engineering (Receivers, Processors, Exporters)	04 • Dark Telemetry Illumination & Latency Bottleneck Detection
05 • Edge PII Scrubbing, Masking & Sensitive Data Governance	06 • Production Incident Triage for 100+ Person Bridge Calls	07 • APM & Backend Integration (OpenObserve, Datadog, ELK, Prometheus)	08 • High-Concurrency Distributed Systems Reliability

ADDITIONAL EXPERIENCE

- **Senior Backend Developer, SK Holdings, Inc.** | January 2019 - December 2020 | Chicago, IL: Led backend stability, performance, and modernization for high-traffic Rails products, improving full-text search, campaign delivery pipelines, and data layer reliability while systems remained under production load.
- **Senior Software Developer, BenchPrep** | March 2017 - February 2018 | Chicago, IL: Owned enterprise assessment workflows, leading correctness and platform security in a high-concurrency environment.
- **Senior Software Developer, ReachLocal** | March 2015 - November 2016 | Remote: Owned API design and modernization strategy, leading incremental legacy migration for a high-volume digital marketing platform.
- **Senior .NET Developer, TicketsNow** | November 2005 - March 2007 | Crystal Lake, IL: Owned real-time inventory systems, leading transactional integrity and iterative delivery for revenue-critical operations.

EARLIER EXPERIENCE

Selected earlier experience

Built and operated enterprise integrations, transactional systems, and early web products across consulting and product organizations, establishing the production discipline and community-centered craftsmanship that inform my leadership today.

- **Revenue-critical commerce:** Designed real-time inventory, locking, and fulfillment services for a high-volume ticket marketplace later acquired by a major industry operator.
- **Consulting and software craftsmanship:** Delivered systems across client environments while mentoring engineers, adopting Ruby and Rails, and helping build Chicago's craftsmanship community.
- **Enterprise and operational systems:** Built integration and operational software across large organizations, grounded in .NET, SQL, messaging, production support, and direct user needs.